TITLE PAGE

- **Problem Statement ID –** 26145
- **Problem Statement Title-**AI-Based Detection of Cyber Threats in Unidirectional IP Traffic
- **Theme-**Blockchain & Cybersecurity
- **PS Category-** Software
- **Team ID-**
- **Team Name (Registered on portal):SAMARAGS**

SAMARAGS

IDEA TITLE

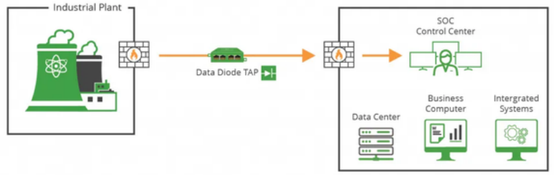
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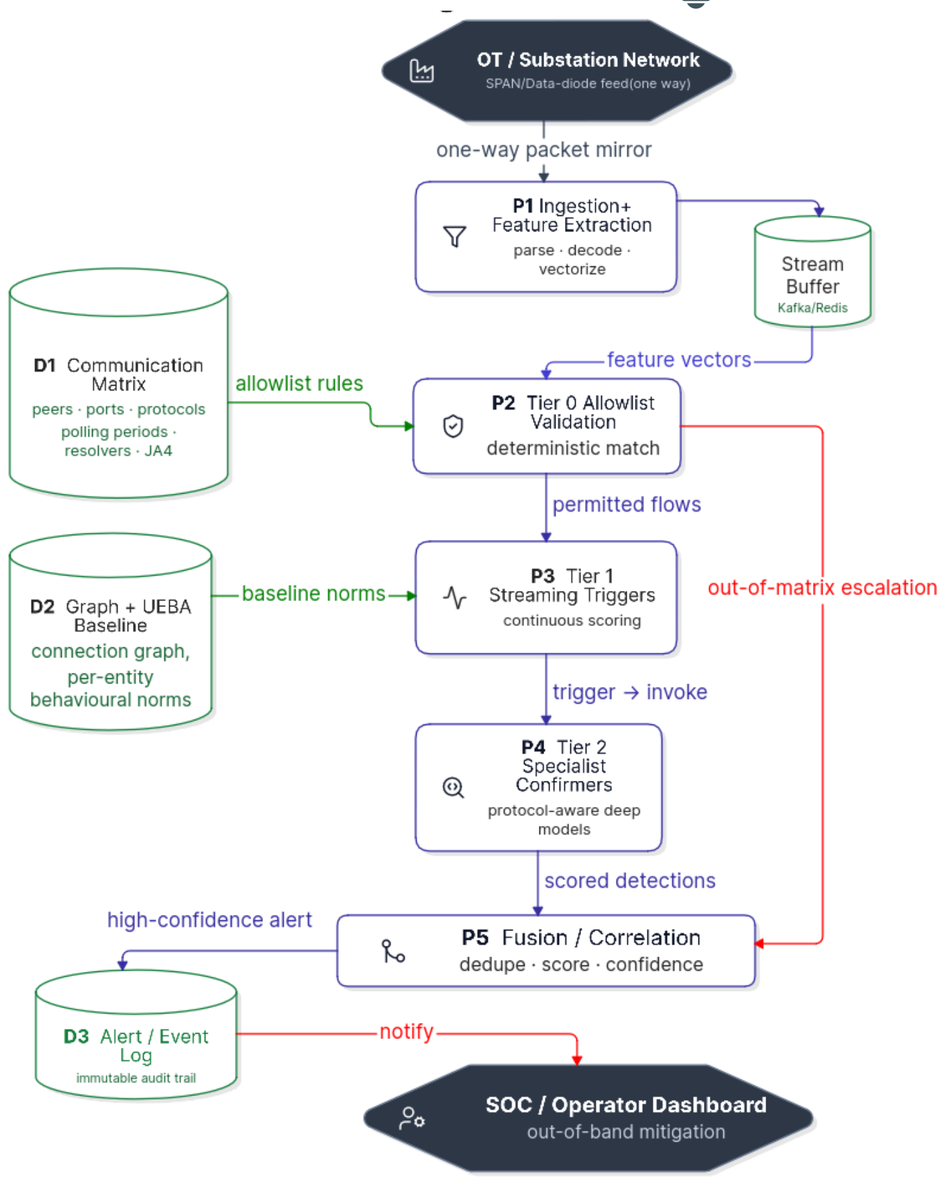
- **Proposed Solution**

The problem — the unidirectional blind spot
Critical infrastructure uses data diodes to keep attackers out. But this creates a blind spot: if the secure network is compromised from the inside (infected USB, rogue insider, supply-chain implant), the malware tries to phone home, spread, or exfiltrate.
No return path — cannot probe, complete handshakes, or block inline.
No decryption — TLS/QUIC seen from metadata only.
Asymmetric, passively observed telemetry (NetFlow / IPFIX / sFlow).

How it addresses the problem
Silently analyses outbound / mirrored streams to catch internal infections a diode alone cannot see.
A single model cannot cover six attack modalities — so a layered ensemble
(allowlist → triggers → confirmers → fusion) covers all of attack types mentioned a–f.
Exploits the fixed OT communication matrix: any off-matrix peer, port, or period is anomalous by construction.
Reconstructs one-way substitutes for lost bidirectional features (entropy, fan-out, JA4 novelty, volume baselines).

Innovation & uniqueness
Purpose-built for the diode constraint — not a repurposed IT IDS.
Comm-matrix allowlist is a first-class detector, not just a filter.
Trigger→confirm tiering: cheap streaming detectors gate heavy models.
Fusion cross-correlates weak signals + maps to MITRE ATT&CK for ICS.
CPU-only, open-source, zero added latency to production flows.





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TECHNICAL APPROACH

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- Technologies to be used (e.g. programming languages, frameworks, hardware)
- Methodology and process for implementation (Flow Charts/Images/ working prototype)

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FEASIBILITY AND VIABILITY

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- Analysis of the feasibility of the idea
- Potential challenges and risks
- Strategies for overcoming these challenges

Samarags

IMPACT AND BENEFITS

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Target sectors & applications (Indian critical infrastructure)

Power Grids & Nuclear (ICS/SCADA)
NTPC · PGCIL · load dispatch centres · CEA guidelines
Defends OT networks pushing live telemetry to IT dashboards; catches reconnaissance and C2 beaconing before physical damage.

Defense & Space (air-gapped telemetry)
ISRO · DRDO · NPCIL · indigenous BEL hardware diodes
Monitors data pushed from high-security enclaves to lower-security stations; tracks asymmetric flow-volume anomalies to stop exfiltration.

Digital Payments & Fintech
NPCI / UPI switches · SWIFT gateways · core banking
Analyses TLS/QUIC (JA3/JA4) metadata for encrypted malware without decrypting sensitive payloads or adding latency.

Smart Manufacturing & Industry 4.0
Reliance Jamnagar · ONGC · IoT sensor fleets
Secures telemetry pushed to cloud platforms; instantly detects spoofed-source floods and botnet takeovers.

Railway & Metro Infrastructure
Indian Railways' Kavach · Delhi Metro signalling
Secures isolated signalling networks from passenger-facing IT, analysing unidirectional log feeds for anomalies.

Benefits

Security: full-spectrum coverage (DDoS → stealth exfil) with one-way, no-decryption visibility.
Zero risk / zero latency: the tap adds nothing to production flows and cannot become a pivot.
Economic: open-source, CPU-only — a fraction of the cost of proprietary detection appliances.
Operational: preserves determinism and safety of the control network.
Social: protects reliable power, payments, transport, and defense telemetry citizens depend on.

Compliance & strategic fit

Aligns with CEA cyber-security guidelines for Indian load dispatch centres.
Complements indigenous BEL hardware diodes in defense / space enclaves.
Chain-of-custody forensic storage supports audit and incident response.
Software-only, no added hardware footprint — sustainable to deploy and scale.

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RESEARCH AND REFERENCES

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- Details / Links of the reference and research work

IMPORTANT INSTRUCTIONS

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Please ensure below pointers are met while submitting the Idea PPT:

1. **Kindly keep the maximum slides limit up to six (6). (Including the title slide)**
2. **Try to avoid paragraphs and post your idea in points /diagrams / Infographics /pictures**
3. **Keep your explanation precise and easy to understand**
4. **Idea should be unique and novel.**
5. **You can only use provided template for making the PPT without changing the idea details pointers (mentioned in previous slides).**
6. **You need to save the file in PDF and upload the same on portal. No PPT, Word Doc or any other format will be supported.**

Note - You can delete this slide (Important Pointers) when you upload the details of your idea on SIH portal.